SoilPulse-ET

SmallSat Onboard Triage for Regenerative Agriculture Water-Stress Follow-Up

Participant Information

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NASA Data Set Used: ECOSTRESS Tiled Evapotranspiration Instantaneous and Daytime L3 Global 70 m V002

DOI: 10.5067/ECOSTRESS/ECO_L3T_JET.002

Public Artifact Repository: https://github.com/SproutSeeds/soilpulse-et
Paper in Repository: docs/PAPER_FINAL_CANDIDATE.pdf

1. Introduction

A. Brief overview of the proposed solution and description of your team/organization.

SoilPulse-ET is a SmallSat onboard triage concept for regenerative agriculture water-stress follow-up. The proposed system keeps a tile-level knowledge ledger, scores event priority and evidence quality, then decides whether each candidate land tile should be sent as a priority image chip, compressed into a feature summary, or deferred during a constrained contact. The submission is prepared by Cody Mitchell as an independent researcher affiliated with Fractal Research Group / SoilPulse-ET. The Phase 1 artifact is a reproducible concept and policy demonstrator: it expresses the triage logic, evidence ledger, and resource accounting in software that reviewers can run, inspect, and extend during future validation work.

B. Condensed summary of how the solution addresses all 5 Judge Criteria concepts.

- Creativity: SoilPulse-ET moves beyond a ground-only dashboard by putting a chip / summary / defer policy onboard the spacecraft planning loop.
- Technical feasibility: the concept is anchored to a conservative 6U CubeSat planning envelope and explicit downlink, packet, processing-time, and energy budgets.
- Potential impact: the current hybrid demo uses 224 KiB versus a 784 KiB full-chip baseline, saving 560 KiB / 71.43% while retaining 2/2 high-stress candidates and deferring 1/1 cloudy candidates.
- Business and market evaluation: the first users are irrigation districts, groundwater agencies, conservation planners, and regenerative agriculture technical-service providers.
- Presentation and reproducibility: the public artifact repository provides runnable policy code, demo fixtures, tests, paper files, and source notes tied to ECOSTRESS L3T JET V002.

2. Proposed Solution Details

A. Explanation of proposed adaptive sensing or onboard processing approach to regenerative agriculture or forestry, or a similar land resilience objective.

SoilPulse-ET supports water-stress triage for regenerative agriculture and related land-resilience workflows. A ground planner nominates candidate field tiles based on user priorities, recent field history, and revisit needs. Onboard processing converts observations into compact features: evapotranspiration anomaly proxy, vegetation-change signal, cloud fraction, days since last useful observation, user priority, confidence, and resource cost. The policy then selects one action per tile: `priority_chip`, `feature_summary`, or `defer`.

B. Problem statement for the solution.

Land managers and SmallSats both face scarcity. On the ground, field teams have limited attention and must decide which water-stress signals deserve follow-up. In orbit, SmallSats have limited power, compute, storage, contact time, and downlink. The core problem is deciding what is worth processing and returning first under constrained mission resources.

I. Selected land use algorithm(s) and small satellite technical requirements.

The first NASA data anchor is ECOSTRESS L3T JET V002. ECOSTRESS supports evapotranspiration logic and future calibration. The current software maps NASA and mission signals into transparent feature fields: ET anomaly proxy, confidence, cloud fraction, revisit age, user priority, and packet-size estimates. The planning envelope is a conservative 6U CubeSat body, with representative onboard processing using a radiation-tolerant CPU for policy execution and optional acceleration for feature extraction.

II. Mission requirements aligned with onboard constraints.

Constraint	SoilPulse-ET design response
Downlink	Select priority chips, send summaries, and defer low-value tiles instead of returning every full chip.
Compute	Use compact per-tile features and a deterministic policy before downlink.
Energy	Track per-tile processing and packet costs under a mission budget.
Storage	Keep a tile-level ledger and return compact evidence when a summary is sufficient.
Contact time	Prioritize packets that produce a ranked tile list for analyst follow-up.

III. Solution design integration into a current commercial market full-stack implementation system.

The concept fits as an onboard decision module for Earth-observation operators and as a downstream input for agricultural analytics providers. A practical implementation path is: user priorities on the ground, spacecraft tasking, onboard feature extraction and triage, downlink of selected packets, and a district or conservation-provider list of tiles for follow-up.

3. Potential Impact

A. Benefits and impact of the proposed solution.

The benefit is fewer low-value packets during a limited contact window. SoilPulse-ET is designed to reduce wasted downlink, avoid returning cloudy or low-confidence observations during scarce contacts, and focus land-resilience analysts on the tiles most likely to need timely follow-up.

I. How this solution builds upon current research and market capabilities.

The project builds on NASA evapotranspiration data, operational interest in satellite-driven agriculture monitoring, and market examples such as ET-informed water budgets and district-scale analytics. Its contribution is a repeatable decision rule and resource ledger for choosing what a SmallSat should sense, summarize, downlink, or defer before data reaches the ground.

II. Effect on technical variables compared to traditional methods.

Variable	Hybrid demo result	Interpretation
Downlink	224 KiB vs. 784 KiB	560 KiB / 71.43% byte savings versus sending all full chips.
Packet count	4 selected packets	Two full chips plus two compact summaries.
Cloud waste	1/1 cloudy deferred	Cloudy evidence is held back instead of consuming downlink.
Stress retention	2/2 retained	The highest-stress candidates remain selected in the fixture.
Energy/processing	38.10% / 48.00%	The policy stays within concept planning budgets.

The 71.43% metric is byte savings. Four of eight tiles are deferred, and two selected tiles use compact summaries, which reduces bytes beyond the tile-count reduction alone.

III. Concrete examples, outperformance scenarios, and limitations.

Example scenario: after a hot and dry week, an irrigation district analyst may need to compare cover-cropped orchards, deficit-irrigated vineyards, rotational pastures, riparian buffers, and managed fallow trials. A traditional full-return workflow can spend downlink on every candidate. SoilPulse-ET instead returns detailed chips for urgent tiles, compact summaries for useful but lower-cost evidence, and defers cloudy or low-confidence tiles. The evidence scope is explicit: the current hybrid fixture contains one ECOSTRESS-derived knowledge-gap row plus labeled synthetic mission-scenario rows. It demonstrates Phase 1 policy behavior and resource accounting, with field validation and flight-readiness testing defined as next-phase work.

4. Team Experience and Market Evaluation

A. Qualifications and expertise of team members.

Cody Mitchell is submitting as an independent researcher affiliated with Fractal Research Group / SoilPulse-ET. The Phase 1 contribution is software systems design, reproducible artifact packaging, and documentation that names the data rows, assumptions, commands, and metrics behind the result. The documented evidence supports triage under downlink, compute, and energy budgets, public review of the artifact, and a list of validation tasks for formal flight qualification, agronomic validation, and future NASA review.

B. Past projects or research relevant to the proposed solution.

Fractal Research Group is used here as the public research umbrella for a small, reviewable artifact. SoilPulse-ET keeps the code, paper, source list, budget table, test fixture, and next-validation plan in one repository so reviewers can see which parts are implemented and which parts are scheduled for follow-up.

C. Analysis for how the proposed solution can eventually be scaled or integrated into future markets.

The initial product would be an analytics and tasking module for Earth-observation operators or agricultural decision-support providers. It can scale from the current runnable fixture to a pilot workflow by adding more real ECOSTRESS windows, real HLS vegetation ingestion, user-calibrated thresholds, and more detailed contact-window models.

I. Potential product users and how target users inform solution design criteria.

Potential users include irrigation district analysts, groundwater sustainability agencies, conservation technical-service providers, regenerative agriculture program managers, and water-risk analytics partners. Their needs drive the design criteria: tile-level reason codes, fewer low-value items in the follow-up queue, cloud and confidence fields on every decision, and predictable behavior when downlink or compute budgets are tight.

II. Potential market applications beyond agricultural or forestry purposes.

The same chip / summary / defer pattern can transfer to forestry drought stress, restoration monitoring, post-fire recovery, riparian-buffer monitoring, wetland resilience, and water-resource risk analytics. The common market need is prioritized evidence under constrained observation and response capacity.

5. Expected Outcomes

A. Development process from base concept to MVP based on market analysis and industry gaps.

The base concept is already demonstrated as a reproducible policy fixture. The MVP path is to harden the evidence base and user fit:

1. Expand ECOSTRESS event-hunt windows across hot, dry, and comparison periods.
2. Replace synthetic HLS and terrain support fixtures with real ingestion.
3. Calibrate chip / summary / defer thresholds with a pilot user persona.
4. Extend the resource model to raw scene sizes, memory, storage, contact windows, and downlink rates.
5. Package a reviewer-friendly dashboard or notebook view for tile decisions and resource budgets.

This process addresses the industry gap between abundant Earth-observation data and the operational need for a ranked, explainable follow-up queue under fixed downlink, compute, and contact-time limits.

B. Next steps if selected as a Finalist.

If selected, the next phase is validation and mission hardening. SoilPulse-ET would test the policy against more real windows, record ambiguous and failed cases, tune thresholds against actual user priorities, and map the concept onto a specific 6U compute, power, thermal, storage, and downlink stack. The finalist target is a better-validated mission prototype with documented evidence around resource budgets, user thresholds, and hardware assumptions.

6. Conclusion

A. Review key points about the proposed adaptive sensing or onboard processing solution.

SoilPulse-ET gives a SmallSat a transparent way to choose which candidate tiles to return first. It maintains a tile-level knowledge ledger, scores event priority and evidence quality, and chooses between priority chips, feature summaries, and deferral. The NASA anchor is ECOSTRESS L3T JET V002, DOI `10.5067/ECOSTRESS/ECO_L3T_JET.002`.

B. Summarize strengths and potential impact.

The strength of SoilPulse-ET is that each tile decision has an action, reason code, packet-size estimate, and budget impact. The current hybrid demo shows meaningful downlink savings while preserving high-stress candidates and deferring cloudy evidence. For regenerative agriculture and water-stress follow-up, this can help missions manage limited satellite resources so land managers can respond faster to scarce water. The Phase 1 package includes runnable policy code, fixture rows, tests, budget metrics, and paper documentation; the next phase adds more real-data windows, user-calibrated thresholds, and a specific 6U compute, power, thermal, storage, and downlink model.

7. References

- NASA Space to Soil Challenge. https://nasa-space-to-soil.org/
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